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Data Provenance Process

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Document Control

This document is the responsibility of the Author. It is subject to formal change control after the initial approved release (e.g. version 1.0). Change requests must be communicated to the Author in writing. Agreed changes are recorded in the Version History of the document.

Version History

Date	Version	Author	Changes
13/04/20XX	0.1	John Doe	New Document – initial draft
01/05/20XX	1.0	John Doe	Issued to Business

Reviewers

Name	Role	Date
Jeremy Doe	Data Management Lead	01/09/20XX

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Process

Background

- Data provenance is the recording of information that describes the origin, history, and context of a dataset. It aids the understanding of how the data was collected, processed, and transformed, and what assumptions, methods, and sources were employed. Data provenance is essential for ensuring the quality, reliability, and trustworthiness of data.
- This procedure operates in parallel with the following processes and procedures:
  - <<reference>> Data Exploration Procedure
  - <<reference>> Data Cleaning Procedure
  - <<reference>> Data Analytics Process
  - <<reference>> Data Enhancement Procedure
  - <<reference>> Data Perturbation Procedure
  - <<reference>> Data Labelling Procedure
  - <<reference>> Data De-identification Procedure
  - <<reference>> Data Normalization and Scaling procedure.
- Data provenance information and records are drawn from all of these procedures.

Provenance Data Model

- Provenance information is recorded in the Data Provenance Database, indexed with a unique provenance reference. The provenance information recorded is as follows:
  - Source
  - Geographical location
  - Collection mechanism
  - Data type
  - Data processing results:
    - Data distribution
    - Statistical analysis information
    - Data duplication
    - Erroneous or missing data
    - Data analysis results
    - Data enhancement / perturbation information
    - Data labelling undertaken
    - De-identification undertaken
    - Normalization and scaling undertaken.

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- Data items are tagged with the relevant provenance reference as they undergo preprocessing.

Process

- Initially, data items received (from production, externally or internally) are examined to determine if similar data items have been preprocessed previously, for which existing data provenance information is held in the Data Provenance Database. If so, the associated provenance reference is recorded for the data item.
- In the event that a previously unprocessed data item is received, the following process steps are completed.
- The overall process flow is set out in Diagram 1 below.

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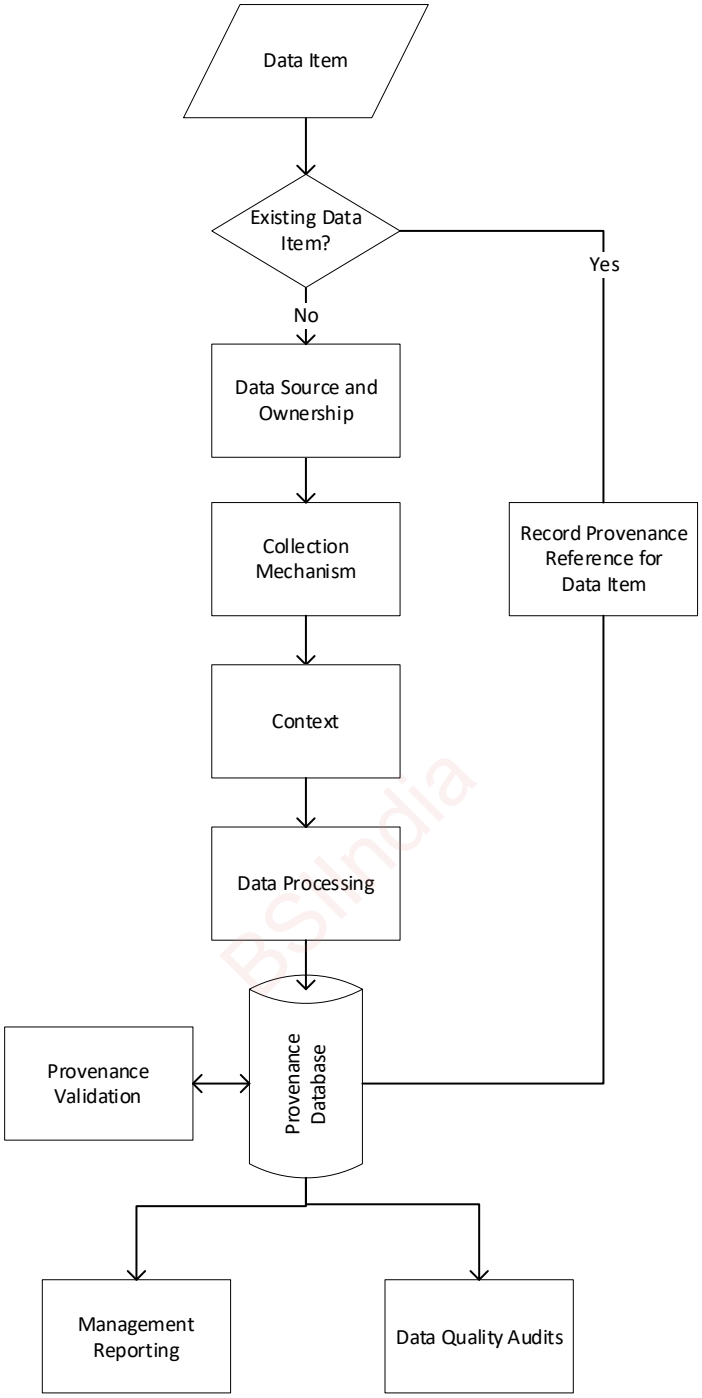

Diagram 1: Data Management Process Flow

Process Steps

Data Source and Ownership

- The source of the data item is established and recorded in the Data Provenance Database, as follows:

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- Internal data: ownership lies with <<organization>>.
- External data: ownership lies with the third party from where the data was sourced.
- Production data: ownership will vary depending on which system the production data was sourced from, the AI consumer (user) that generated the data and the AI subject (which may be the user, or another individual).

- Where production data relates to an AI subject with whom there is no direct relationship, source information may need to be obtained from the user.
- If it is necessary to seek information from a user, this needs to be undertaken through the relevant customer account manager.
- Note that none of the <<organization>> AI systems draw data from public domain web sources. As such, issues relating to the identification of ownership of the data sourced from the web do not apply.

Collection Mechanism

- Record details of the mechanism by which the data item was collected, including:
  - Real-time feed: application programming interface (API), middleware or database transfers, details of security measures and/or encryption employed.
  - Overnight / batch feed: file transfer approach, including intermediary servers/services used, by whom the file transfer process is operated, security mechanisms in place, cloud services employed.
  - Manual: user interface through which data is manually keyed / copied, details of any tools involved.

Context

- Record details of the following:
  - Geographical region from where the data was obtained.
  - Applicable regulatory and legislative jurisdiction.
  - Market, industry or sector applicable.
  - Applicable privacy legislation.

Data Processing

- Drawing on records from the exploration, cleaning, analytics, enhancement, perturbation, labelling / de-identification and normalization and scaling procedures, record details of any changes that have been made to or affected the data item.

Provenance Validation

- Once all provenance information has been recorded in the Data Provenance Database, the provenance information is validated using the ProvValidator tool. The tool helps to identify any missing information, as well as duplication of provenance information within the database.

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- If duplication is identified, the provenance reference for the data item must be updated to the original Data Provenance Database reference, and the duplicate record removed from the Data Provenance Database.
- Through this approach, the integrity of the Data Provenance Database is subject to continuous review.

Management Reporting

- Provenance information is included in management reports produced for the Senior Leadership Team. The provenance landscape is visualized using ProvVis, with summary information included in management reports, highlighting any shifts in data provenance and in particular any new applicable:
  - Legislative jurisdictions.
  - Regulatory jurisdictions.
  - Privacy legislation.

Data Quality Audits

- Data provenance information is used as part of data quality audits and in doing so is subject to verification as part of these audits.
- The Quality Assurance Team has access to the Data Provenance Database for the conduct of periodic data quality audits. In addition, the Data Management Team will make available any supporting information to the Quality Assurance Team upon request.

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